FIKR Model — Lesson Plan

Learning Leading to Production

Title	Overview of Complex Numbers	**Subject / Unit**	Topic 2 — Quadratic Functions and Equations · Lesson 2-4
Grade	Grade 10 — Algebra II	**Date**	Week 7 · Semester 1, 2026–27
Format	[ ] 40-minute [X] 60-minute	**Teacher**	Mr Malek Thiab

Essential Question (curriculum map)

Not shown — the curriculum map's Essential Question column for Topic 2 was not in the project text, so it is left open rather than invented. Add it when confirmed.

Lesson Objectives (curriculum map — verbatim)

1. Demonstrate the role of the imaginary unit i in the system of complex numbers.
2. Evaluate expressions involving the imaginary unit i and complex numbers.
3. Use the properties of complex numbers to perform operations such as addition, subtraction, and multiplication.
4. Use complex conjugates to simplify quotients of complex numbers.
5. Solve real-world problems using complex conjugates.
6. Express the complex solutions of a quadratic equation in the form a + bi.

Vocabulary (curriculum map) · Assessments (curriculum map)

Vocabulary: Argument ; complex conjugate ; complex number ; complex plane ; imaginary axis ; imaginary part ; imaginary unit ; Modulus ; Norm ; purely imaginary number ; real axis ; real part.

Assessments: “Give it a go” questions during practice ; “Time to Check” at the Mastery Gate.

0 Preparation Before Class

Learning Outcome: What will the student be able to do?

Students will use the imaginary unit i to simplify square roots of negative numbers and to evaluate powers of i (objectives 1 and 2; HSN.CN.A.1).

Students will add, subtract and multiply complex numbers and write every answer in the form $a + bi$ (objective 3; HSN.CN.A.2).

Students will use the conjugate to find $z\bar{z}$ and the modulus, to divide complex numbers, and to solve a model circuit problem (objectives 4 and 5; HSN.CN.A.3).

Students will express the complex solutions of a quadratic equation in the form $a + bi$ (objective 6; HSA.REI.B.4.B).

The curriculum map’s Essential Question and Math Practices columns for Topic 2 were not in the project text; they are left open, not invented.

Three teaching days this week. If time runs short, cut Smart Production and carry it into the next session — never the Mastery Gate, which is the evidence.

Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?

Flipped video (4 min): "What is i?" — why we invent a number whose square is -1. Posted on the LMS 48 hours ahead.

Two embedded EdPuzzle questions: simplify $\sqrt{-16}$, and write the value of i^2.

Evidence of readiness: EdPuzzle completion report plus both embedded answers submitted before the bell.

1 Diagnostic 6 min

Tool Used: What evidence indicates differences in students' current levels?

3-question Quizizz (≤ 4 minutes) — the slide-4 checks: simplify $\sqrt{72}$; expand $(x+3)(x-2)$; find the discriminant of x^2+2x+5.

Answers: $6\sqrt{2}$; x^2+x-6 ; -16. No calculator.

Where is the learning gap? What will you adjust in your instruction accordingly?

Expected gap 1 — Q1: students stop at $2\sqrt{18}$ because they do not take out the largest square factor; complex numbers need clean radicals.

Expected gap 2 — Q2: the middle term is forgotten; multiplying $a+bi$ by $c+di$ uses exactly the same four products.

Routing: 0–1 correct → re-teach square roots and expanding two brackets with the teacher. 2 → straight to i. 3 → move quickly through block 1 and begin the Investigate prompt early.

2 Targeted Instruction 18 min

Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?

Slide 6 — The imaginary unit (objectives 1 and 2; HSN.CN.A.1). $i=\sqrt{-1}$, $i^2=-1$. Examples: $\sqrt{-49}=7i$; $i^{27}=-i$ because $27=4\cdot 6+3$; $\sqrt{-4}\cdot\sqrt{-9}=-6$.

Slide 7 — Add, subtract and multiply (objective 3; HSN.CN.A.2). $(a+bi)\pm(c+di)=(a\pm c)+(b\pm d)i$. Examples: $(3+2i)+(5-6i)=8-4i$; $(2+3i)(4-i)=11+10i$.

Slide 8 — Conjugates, modulus and the complex plane (objective 4; HSN.CN.A.3). $\overline{a+bi}=a-bi$, $(a+bi)(a-bi)=a^2+b^2$, $|a+bi|=\sqrt{a^2+b^2}$. Examples: $|3+4i|=5$; $x^2+9=(x+3i)(x-3i)$.

Slide 9 — Dividing and complex solutions (objectives 4 and 6; HSA.REI.B.4.B). Multiply top and bottom by the conjugate. Examples: $\dfrac{3+i}{1+i}=2-i$; $x^2-4x+13=0$ gives $x=2\pm 3i$.

Narration focus — take i out FIRST: $\sqrt{-4}\cdot\sqrt{-9}$ is $2i\cdot 3i=-6$, not $\sqrt{36}$.

Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?

On whiteboards, 60 seconds: simplify $\sqrt{-25}$ and evaluate i^6.
Expected: $5i$ and -1. Watch for -5 (dropping the i) and $i^6 = 6$ or i.
80% correct → release guided practice. Below 80% → one more modelled example with a different exponent.

3 Practice 12 min

Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?

Two "Give it a go" problems, identical for every student, worked together with the teacher:
(1) Simplify $\sqrt{-75}$. (2) Multiply $(3-2i)(1+5i)$.
Answers: (1) $5\sqrt{3}\,i$. (2) $3+15i-2i-10i^2 = 13+13i$.
Feedback: worked live on the board step by step; students self-mark and circle their OWN first error.

Independent Task and Feedback Tool: students choose their route and may switch at any time

Delivered through the "Choose Your Route" sheet with live feedback in Pear Deck. Routes are named for the task, not for the student.

PRACTICE — Secure the method. Simplify $\sqrt{-18}$, evaluate i^{13}, add and multiply complex numbers, divide $\dfrac{8-i}{2+i}$ and solve $x^2+2x+5=0$. Done when: i^2 is replaced by -1 in every product and every answer is in the form $a+bi$.

APPLY — Use it in context. Add two impedances, find the current $I = \dfrac{V}{Z}$ and its modulus, solve $x^2-10x+29=0$, and find $a+bi$ with $(a+bi)^2 = -5+12i$. Done when: the quotient is checked by multiplying back and the solutions are checked by substitution.

INVESTIGATE — Find out why. Why $(a+bi)(a-bi)$ is always real; whether $\sqrt{-4}\cdot\sqrt{-9} = \sqrt{36}$; why the two solutions of a real quadratic with $b^2-4ac<0$ are conjugates. Done when: each argument uses $i^2=-1$ in general terms, not only one numerical example.

Every student sits the same Mastery Gate. The gate is not differentiated, because it is the evidence.

4 Production — Stage One 12 min

Production Task: What new task, situation, or product will allow the student to demonstrate learning?

Non-routine transfer task — this context has not appeared in practice (slide 11).
A model circuit at a NEOM coastal desalination plant (a Vision 2030 project on the Red Sea): two parts in series have impedances $Z_1 = 2+3i$ and $Z_2 = 1+i$ ohms, and the supply is $V = 120$ V with $I = \dfrac{V}{Z_1+Z_2}$. All numbers are model values, not plant data.
(a) Add the impedances to find Z. (b) Write the current I in the form $a+bi$ using the conjugate. (c) Find the

modulus of I and decide whether a 30 A breaker is enough.

Answers: (a) $Z = 3 + 4i$ ohms. (b) $I = \frac{120(3-4i)}{25} = 14.4 - 19.2i$ A. (c) $|I| = \sqrt{14.4^2 + 19.2^2} = 24$ A, which is less than 30 A, so the breaker is enough in this model (check: $|I| = \frac{120}{|Z|} = \frac{120}{5} = 24$).

Students then hand their reasoning to the AI critic.

Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?

Minimum acceptable standard: Z found by adding real and imaginary parts separately, the current written as $a + bi$ after multiplying by the conjugate, and units (ohms, amperes) in every answer.

Role of AI: critic only. Briefed prompt — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently by the student: the written justification for parts (a)–(c). AI may challenge it; it may not write it.

5 Presentation — Evidence of Learning 6 min

Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?

"Time to Check" — one individual task. No notes, no partner, no AI.

1. Evaluate i^{50}. 2. Multiply $(2+i)(3-2i)$. 3. Simplify $\frac{4}{1+i}$.

Answers: 1. $i^{50} = i^2 = -1$. 2. $6 - 4i + 3i - 2i^2 = 8 - i$. 3. $\frac{4(1-i)}{2} = 2 - 2i$.

Done when: each answer is in the form $a + bi$ and question 3 shows the conjugate.

Scored live in Pear Deck as submissions arrive; result logged in the Impact Dashboard Mastery Rate.

6 Final Product — Refinement 6 min

Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?

Students revise their circuit reasoning using the AI critique and the teacher's micro-conference note.

Final product, three equally acceptable formats: a Canva one-pager, a photographed hand-written solution, or a 45-second voice note.

Required content: Z, the current in the form $a + bi$, its modulus checked as $\frac{120}{|Z|}$; ONE worked non-example — writing $\sqrt{-4} \cdot \sqrt{-9}$ as $\sqrt{36} = 6$ — with a sentence on why i must be taken out first; and a caption on what the modulus means for the breaker.

Graded on originality and clarity of reasoning, not correctness alone.

Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall as the Topic 2 modelling set.

Reflection question: "Which is harder for you — remembering that $i^2 = -1$ inside a product, or dividing with the conjugate?"

Total minutes across the seven stages: 60 (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

★ Required Pillars

Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. Simplify, evaluate powers, add, multiply, divide and solve. Done when: every answer is in the form $a + bi$.

APPLY — Use it in context. A series circuit, a quadratic with complex roots and a square root of a complex number. Done when: the quotient and the roots are checked.

INVESTIGATE — Find out why. Why $(a + bi)(a - bi)$ is real, a counter-example to $\sqrt{a}\sqrt{b} = \sqrt{ab}$, and conjugate roots. Done when: each argument uses $i^2 = -1$ in general terms.

No student is labelled. The routes describe tasks; every student sits the same Mastery Gate.

Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

Saudi Culture & Real-World Connection

Production and project context: a model series circuit feeding a seawater pump at a NEOM coastal desalination plant on the Red Sea (invented model values), a Vision 2030 water-security setting.

Discussion prompt: why do engineers use complex numbers for alternating current instead of only real numbers?

SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAT — Advanced Math: if $\dfrac{7 + 4i}{2 - i} = a + bi$, find b. Multiply by the conjugate $2 + i$: $2 + 3i$, so $b = 3$. Trap: dividing real and imaginary parts separately.

GAT (Qudurat) — which is greater, i^{2026} or i^{2028}? $i^{2026} = i^2 = -1$ and $i^{2028} = 1$, so the second.

SAAT (Tahsili) — the solutions of $x^2 - 2x + 5 = 0$ are $1 \pm 2i$. Trap: stopping at $\sqrt{-16}$ or writing no solution.

Suggestions for Improvement: